

Notes: Diether, Malloy, and Scherbina (JF, 2002)

Differences of Opinion and the Cross Section of Stock Returns

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1 Big picture

- **Question.** Do differences of opinion about a stock predict future returns? The paper studies whether dispersion in analysts' earnings forecasts forecasts low future returns, high future returns, or no future return pattern.
- **Core empirical result.** Stocks with high forecast dispersion earn lower future returns than otherwise similar stocks. The baseline low-minus-high dispersion portfolio earns about 0.79 percent per month, or about 9.48 percent per year.
- **Interpretation.** Forecast dispersion is interpreted as a proxy for disagreement among investors. When pessimistic investors do not trade, prices can reflect the views of optimists, so high-disagreement stocks become overpriced and subsequently earn low returns.
- **Key cross-sectional pattern.** The effect is strongest in small stocks and in stocks that performed poorly over the previous year. These are settings in which short-sale constraints, information frictions, or limits to arbitrage are likely to matter more.
- **Risk story rejected.** Dispersion is positively related to common risk proxies such as beta, return volatility, and earnings variability. If dispersion simply measured risk, higher-dispersion stocks should earn higher returns. The paper finds the opposite.
- **Main economic message.** Investors should not treat forecast disagreement as compensated risk. In the data, disagreement appears to identify stocks whose prices are too optimistic relative to future payoffs.

2 Incremental contribution of the paper

- **Direct empirical test of the Miller disagreement idea.** The paper gives large-sample evidence for the Miller (1977) hypothesis that when pessimists are constrained from trading, prices reflect optimistic beliefs and future returns are lower.
- **Forecast dispersion as a market-wide disagreement proxy.** Instead of relying only on ownership breadth or trading volume, the paper uses dispersion in analysts' earnings forecasts as a direct measure of disagreement about firm fundamentals.
- **Large cross-sectional evidence.** The sample combines CRSP returns, Compustat accounting data, and I/B/E/S analyst forecasts over 1983-2000. The study therefore moves beyond earlier small-sample evidence on analyst disagreement.
- **Data-cleaning contribution.** The authors identify a problem with split-adjusted I/B/E/S data: rounding after historical split adjustments can mechanically create false zero dispersion. They therefore use raw forecast data.
- **Risk-versus-disagreement separation.** The paper shows that dispersion correlates with risk proxies, but the return premium has the wrong sign for a standard risk interpretation and survives multifactor adjustment.
- **Robustness contribution.** The return effect survives size, book-to-market, momentum, turnover, analyst coverage, forecast age, and Fama-MacBeth controls. It is also visible under different lags and holding periods.

- **Alternative friction.** Beyond formal short-sale constraints, the paper argues that analyst self-censoring can prevent negative opinions from reaching prices. Pessimistic analysts may stop covering a stock or avoid issuing very low forecasts, creating optimistic consensus forecasts.

3 Data and measurement

3.1 Forecast and returns data

- Returns are from CRSP for NYSE, AMEX, and Nasdaq stocks.
- Accounting variables such as book equity are from Compustat.
- Analyst earnings forecasts are from I/B/E/S.
- The main portfolio tests use February 1983 through December 2000.
- A stock must have at least two analysts' forecasts and a price above five dollars.

Interpretation: The sample is not the entire CRSP universe. It is the universe of stocks with enough analyst coverage to measure disagreement. This matters because the eligible sample is tilted toward larger firms.

3.2 Dispersion measure

The main disagreement proxy is

$$Dispersion_{i,t} = \frac{\sigma(\text{analysts' EPS forecasts}_{i,t})}{|\overline{\text{EPS forecast}}_{i,t}|}.$$

- The numerator is the standard deviation across analysts' current-fiscal-year EPS forecasts.
- The denominator scales dispersion by the absolute value of the mean forecast.
- If the mean forecast is zero, the observation is assigned to the highest dispersion group.
- Alternative scaling by book equity per share gives similar results.

Interpretation: The variable is designed to capture disagreement relative to the level of expected earnings. It is not simply the dollar size of forecast disagreement.

3.3 Portfolio construction

- Each month, stocks are assigned to dispersion portfolios using the previous month's forecasts.
- Baseline portfolios are held for one month.
- Portfolio returns are equal weighted.
- Many tests use conditional sorts by size, book-to-market, and momentum.

Interpretation: Equal weighting gives substantial influence to smaller stocks, which is important because the disagreement effect is strongest among smaller firms.

3.4 Additional variables

The paper also uses beta, size, book-to-market, past returns, residual analyst coverage, return volatility, trading volume, turnover, leverage, sales-to-assets, earnings variability, and forecast age.

Interpretation: These variables help distinguish the disagreement explanation from risk, liquidity, coverage, size, value, and momentum explanations.

4 Models and empirical specifications

4.1 Main portfolio test

Let $D1$ denote the low-dispersion portfolio and $D5$ the high-dispersion portfolio. The main portfolio spread is

$$Spread_t = R_{D1,t} - R_{D5,t}.$$

- Under the price-optimism/disagreement hypothesis, $E[Spread_t] > 0$.
- Under the unbiased-price hypothesis, $E[Spread_t] = 0$.
- Under a risk-proxy interpretation, if dispersion measures risk, $E[Spread_t] < 0$.

Interpretation: The sign of the spread is central. A positive low-minus-high dispersion return is evidence that high-disagreement stocks are overpriced, not merely risky.

4.2 Multifactor time-series regressions

The paper estimates three- and four-factor regressions of the form

$$R_{p,t} - R_{f,t} = \alpha_p + b_p(R_{M,t} - R_{f,t}) + s_pSMB_t + h_pHML_t + m_pUMD_t + \varepsilon_{p,t}.$$

- SMB captures the size factor.
- HML captures the value factor.
- UMD captures momentum.
- The key object is the alpha on the high-dispersion portfolio.

Interpretation: If standard risk factors explain the return pattern, high-dispersion alphas should be close to zero. They are not.

4.3 Fama-MacBeth regressions explaining dispersion

The paper estimates yearly cross-sectional regressions of the form

$$Dispersion_i = a_t + \Gamma'_t X_i + u_i.$$

- X_i includes risk proxies, size, value, momentum, coverage, turnover, leverage, sales, earnings variability, and forecast age.
- The goal is to understand what kind of firms have high dispersion.

Interpretation: Dispersion is higher for volatile and hard-to-value stocks, but this does not make it a compensated-risk variable in returns.

4.4 Fama-MacBeth return regressions

The cross-sectional return regressions have the conceptual form

$$R_{i,t+1} = a_t + b_t \text{Dispersion}_{i,t} + \Theta'_t \text{Controls}_{i,t} + e_{i,t+1}.$$

- Controls include size, book-to-market, past returns, residual coverage, turnover, lagged dispersion, lagged standard deviation of forecasts, and mean forecast.
- The main test is whether b_t remains negative after controlling for known return predictors.

Interpretation: The dispersion effect is not mechanically generated by size, value, momentum, short-term reversal, long-run reversal, analyst coverage, or trading volume.

4.5 Analyst optimism and forecast errors

The paper also estimates

$$FE_{i,t} = a_0 + a_1 \text{Dispersion}_{i,t} + \varepsilon_{i,t},$$

where forecast error is realized EPS minus forecasted EPS.

Interpretation: Since average forecast errors are negative, analysts are optimistic on average. A negative a_1 implies that analyst optimism is more severe when forecast dispersion is high.

5 Identification and empirical design

5.1 Three competing hypotheses

- **Disagreement plus constrained pessimists:** high dispersion predicts low future returns.
- **Unbiased prices with fully rational market makers/arbitrageurs:** dispersion does not predict returns.
- **Risk proxy:** high dispersion predicts high future returns.

Interpretation: The empirical design is essentially a horse race among these hypotheses.

5.2 What the evidence can identify

- The paper identifies a robust predictive relation between forecast dispersion and returns.
- It does not randomly assign disagreement across firms.
- The evidence is strongest as a rejection of the risk-proxy interpretation and as support for price optimism under limits to arbitrage.

5.3 Why the cross-sectional patterns matter

- The effect is stronger in small stocks, where arbitrage is harder and shorting may be more costly.
- The effect is stronger among past losers, where negative information and pessimistic views are more likely to be important.
- The effect is not eliminated by turnover or analyst coverage, which weakens alternative slow-information-diffusion explanations.

5.4 Limitations

- Analyst forecast dispersion is an imperfect proxy for investor disagreement.
- The paper does not directly observe all short-sale costs or all investors' beliefs.
- I/B/E/S coverage excludes many very small and distressed firms.
- The return effect weakens in the later subperiod, suggesting that market structure, trading costs, or information technology can affect the strength of the anomaly.

6 Tables and figures

6.1 Table I: Summary statistics on analyst coverage

Read:

- Table I shows that analyst coverage expands over time. In January 1976, only 12.8 percent of CRSP stocks are eligible; by December 2000, the eligible share is 40.5 percent.
- Eligible stocks are much larger than the average CRSP stock.
- In January 1983, only 4.1 percent of stocks in the smallest NYSE decile are eligible, while more than 90 percent of stocks in the largest deciles are eligible.
- The mean number of estimates also rises strongly with firm size.

Economic message: The I/B/E/S sample is not a random sample of all stocks. It is tilted toward firms with analyst coverage, especially larger firms. This makes it notable that the strongest return effects still appear among the smaller stocks that do enter the covered sample.

6.2 Table II: Mean portfolio returns by size and dispersion

Read:

- Across all stocks, the low-dispersion portfolio earns 1.48 percent per month, while the high-dispersion portfolio earns 0.69 percent.
- The low-minus-high spread is 0.79 percent per month with a t -statistic of 2.88.
- The effect is strongest in the smallest size quintile: 1.37 percent per month with a t -statistic of 5.98.

- The spread declines monotonically with size and is small among the largest stocks.

Economic message: Disagreement predicts lower subsequent returns, especially where trading frictions and limits to arbitrage are likely strongest.

Table I
Summary Statistics on Analyst Coverage:
January 1976 to December 2000

Descriptive statistics for NYSE, AMEX, and Nasdaq stocks during the period from January 1976 to December 2000. Panel A reports statistics for all stocks, while Panel B limits the sample to only eligible stocks. A stock is "eligible" to be included in our analysis if it has a one (fiscal) year I/B/E/S earnings estimate, is covered by two or more analysts, and has a price greater than five dollars. Panels C and D report the same statistics as in Panels A and B for January 1983, and breaks the sample down by size decile (formed using NYSE-based market capitalization deciles).

Summary Statistics for 1976–2000						
Panel A: All CRSP Stocks				Panel B: Eligible Stocks		
Date	Number of Firms	Mean Size (Millions)	Percentage of Firms Eligible	Number of Firms	Mean Size (Millions)	Mean No. of Estimates
01/1976	5,065	184.6	12.8	650	910.2	6.47
12/1979	4,781	267.5	29.2	1,395	621.5	7.47
12/1982	5,438	305.6	31.9	1,735	766.0	8.78
12/1985	6,227	362.4	35.4	2,203	938.5	9.95
12/1988	6,798	408.6	34.0	2,309	1,094.1	9.82
12/1991	6,643	610.1	35.5	2,361	1,590.1	9.23
12/1994	8,029	633.8	40.7	3,269	1,423.9	8.26
12/1997	8,839	1,241.2	46.8	4,140	2,473.1	7.19
12/2000	7,823	2,032.9	40.5	3,166	4,740.6	7.79

Summary Statistics by Size for 01/1983						
Panel C: All CRSP Stocks				Panel D: Eligible Stocks		
NYSE Breakpoints	Number of Firms	Mean Size (Millions)	Percentage of Firms Eligible	Number of Firms	Mean Size (Millions)	Mean No. of Estimates
Decile 1	2,969	15.2	4.1	123	34.1	2.76
Decile 2	517	61.2	31.9	165	62.7	3.39
Decile 3	406	96.9	50.5	205	97.8	3.81
Decile 4	323	151.5	69.3	224	151.0	4.90
Decile 5	258	229.4	79.1	204	229.2	6.79
Decile 6	231	343.3	84.0	194	343.9	8.26
Decile 7	191	528.5	89.0	170	526.8	11.22
Decile 8	173	809.1	89.0	154	809.9	13.60
Decile 9	160	1,381.2	92.5	148	1,366.2	16.91
Decile 10	165	5,709.0	92.1	152	5,079.4	19.59

(a) Table I: analyst coverage and eligible stocks

Table II
Mean Portfolio Returns by Size and Dispersion
in Analysts' Forecasts

Each month stocks are sorted in five groups based on the level of market capitalization as of the third Thursday of the previous month. Stocks in each size group are then sorted into five additional groups based on dispersion in analyst earnings forecasts for the previous month. Dispersion is defined as the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast, as reported in the I/B/E/S Summary History file. Stocks with a mean forecast of zero are assigned to the highest dispersion groups, and stocks with a price less than five dollars are excluded from the sample. Stocks are held for one month, and portfolio returns are equal-weighted. The time period considered is February 1983 through December 2000. The table reports average monthly portfolio returns; *t*-statistics in parentheses are adjusted for autocorrelation.

Mean Returns						
Dispersion Quintiles	Size Quintiles					All Stocks
	Small		Large			
	S1	S2	S3	S4	S5	
D1 (low)	1.52	1.45	1.50	1.51	1.48	1.48
D2	1.12	1.40	1.41	1.18	1.35	1.36
D3	0.99	1.20	1.32	1.11	1.36	1.23
D4	0.76	1.07	1.18	1.33	1.33	1.12
D5 (high)	0.14	0.56	0.83	1.03	1.20	0.69
D1–D5	1.37*	0.89*	0.67 ^b	0.48	0.29	0.79*
<i>t</i> -statistic	(5.98)	(3.12)	(2.41)	(1.55)	(0.94)	(2.88)

Mean Dispersion						
Dispersion Quintiles	Size Quintiles					All Stocks
	Small		Large			
	S1	S2	S3	S4	S5	
D1 (low)	0.010	0.011	0.012	0.014	0.014	0.012
D2	0.039	0.033	0.030	0.028	0.025	0.030
D3	0.081	0.062	0.053	0.047	0.039	0.053
D4	0.172	0.125	0.103	0.086	0.067	0.105
D5 (high)	1.256	0.963	0.813	0.722	0.462	0.852

^{a,b} Statistically significant at the one and five percent levels, respectively.

(b) Table II: size and dispersion portfolio returns

6.3 Table III: NYSE market capitalization deciles

Read:

- Table III repeats the dispersion sort within NYSE-based size deciles.
- The low-minus-high spread remains positive in all reported deciles.
- The spread is statistically significant in several middle-to-large NYSE deciles, including deciles 6 through 9.
- The largest NYSE decile has a positive but statistically insignificant spread.

Economic message: The result is not just an artifact of using I/B/E/S size quintiles. Disagreement predicts returns across a broad range of firm sizes, though the effect is still weaker among the very largest stocks.

6.4 Table IV: Size, book-to-market, and dispersion

Read:

- Table IV controls jointly for size and book-to-market by sorting on all three variables.

- The dispersion spread is still significant in several cells, especially among small stocks.
- There is no simple book-to-market pattern that explains the effect.
- Value stocks tend to have higher mean dispersion than growth stocks.

Economic message: The dispersion effect is not the value premium in disguise. Even within size and book-to-market groups, high-dispersion stocks tend to earn lower returns.

Table III

Mean Portfolio Returns by NYSE Market Capitalization Deciles

Each month stocks in each NYSE-based market capitalization decile (which is determined as of the third Thursday of the previous month) are sorted into five groups based on dispersion in analyst earnings forecasts for the previous month. Dispersion is defined as the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast, as reported in the I/B/E/S Summary History file. Stocks with a mean forecast of zero are assigned to the highest dispersion groups, and stocks with a price less than five dollars are excluded from the sample. Stocks are held for one month, and portfolio returns are equal-weighted. Only portfolios in the largest seven deciles are reported since there are fewer than eight stocks on average within each size/dispersion portfolio for size deciles 1 to 3. The time period considered is February 1983 through December 2000. The table reports average monthly portfolio returns; *t*-statistics in parentheses are adjusted for autocorrelation.

Mean Returns							
NYSE Market Capitalization Deciles (Average Number of Stocks in Each Portfolio)							
Dispersion Quintiles	4 (27)	5 (47)	6 (68)	7 (91)	8 (119)	9 (149)	10 (189)
<i>D1</i> (low)	1.21	1.22	1.34	1.42	1.50	1.47	1.45
<i>D2</i>	1.41	1.51	1.26	1.39	1.38	1.18	1.31
<i>D3</i>	1.38	1.31	1.41	1.04	1.30	1.16	1.20
<i>D4</i>	0.72	0.91	0.96	1.11	0.88	1.08	1.18
<i>D5</i> (high)	0.61	0.53	0.33	0.32	0.49	0.64	1.03
<i>D1-D5</i>	0.60	0.69	1.01 ^a	1.09 ^a	1.01 ^a	0.84 ^b	0.41
<i>t</i> -statistic	(1.48)	(1.86)	(3.08)	(3.61)	(3.93)	(2.33)	(1.49)

^{a,b} Statistically significant at the one and five percent levels, respectively.

(a) Table III: NYSE size deciles

Table IV

Mean Portfolio Returns by Size, Book-to-Market, and Dispersion

Each month stocks are sorted into three groups based on the level of market capitalization at the end of the previous month. Each size group is then sorted into three book-to-market groups. The book-to-market ratio is computed by matching the yearly *BE* figure for all fiscal years ending in calendar year $t - 1$ to returns starting in July of year t ; this figure is then divided by market capitalization at month $t - 1$ to form the book-to-market ratio, so that the book-to-market ratio is updated each month. Each size and book-to-market group is further sorted into three dispersion groups. Dispersion is defined as the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast, as reported in the I/B/E/S Summary History file. Stocks with a mean forecast of zero are assigned to the highest dispersion groups, and stocks with a price less than five dollars are excluded from the sample. Stocks are held for one month, and portfolio returns are equal-weighted. The sample period is February 1983 through December 2000; *t*-statistics in parentheses are adjusted for autocorrelation.

Mean Returns								
Dispersion	Low Book-to-Market			Medium Book-to-Market			High Book-to-Market	
	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap
Low	1.35	1.42	1.40	1.64	1.50	1.32	1.50	1.58
Medium	1.27	1.40	1.20	1.29	1.11	1.27	1.13	1.46
High	0.73	1.28	1.38	0.67	0.86	1.12	0.70	1.10
Low-high	0.63 ^b	0.14	0.02	0.96 ^a	0.64 ^b	0.21	0.80 ^a	0.48
<i>t</i> -statistic	(2.10)	(0.47)	(0.06)	(4.09)	(2.37)	(0.96)	(4.09)	(1.70)
Mean Dispersion								
Dispersion	Low Book-to-Market			Medium Book-to-Market			High Book-to-Market	
	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap
Low	0.04	0.03	0.02	0.05	0.03	0.02	0.10	0.04
Medium	0.08	0.05	0.03	0.09	0.05	0.04	0.21	0.11
High	0.49	0.34	0.20	0.56	0.37	0.26	1.04	0.86

^{a,b} Statistically significant at the one and five percent levels, respectively.

(b) Table IV: size, book-to-market, and dispersion

6.5 Table V: Size, momentum, and dispersion

Read:

- Table V controls jointly for size and past returns.
- The strongest effect is among small past losers: the low-minus-high dispersion spread is 0.82 percent per month with a *t*-statistic of 3.65.
- The effect is also significant among small stocks in the middle momentum group.
- Among winners, the dispersion spread is weak or absent.

Economic message: Disagreement is most predictive when the stock has recently done poorly. This is consistent with negative opinions being especially important and difficult to impound into prices.

6.6 Table VI: Three- and four-factor risk adjustment

Read:

- The high-dispersion portfolio has a three-factor alpha of -0.58 percent per month and a four-factor alpha of -0.35 percent per month.
- Both high-dispersion alphas are statistically negative.
- Adding momentum reduces the magnitude but does not eliminate the abnormal low return.
- The GRS tests reject the hypothesis that the factor models explain the dispersion-portfolio average returns.

Economic message: Standard asset-pricing factors cannot explain why high-dispersion stocks underperform. The anomaly is not simply size, value, or momentum exposure.

Table V
Mean Portfolio Returns by Size, Momentum, and Dispersion

Each month stocks are sorted into three groups based on the level of market capitalization at the end of the previous month. Each size group is then sorted into three momentum groups; momentum is computed based on past returns from $t - 12$ to $t - 2$, where "Losers" is an equal-weighted portfolio of stocks in the *worst*-performing 33 percent. Each size and momentum group is further sorted into three dispersion groups. Dispersion is defined as the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast, as reported in the I/B/E/S Summary History file. Stocks with a mean forecast of zero are assigned to the highest dispersion groups, and stocks with a price less than five dollars are excluded from the sample. Stocks are held for one month, and portfolio returns are equal-weighted. The sample period is February 1983 through December 2000; t -statistics in parentheses are adjusted for autocorrelation.

Mean Returns									
Dispersion	Losers						Winners		
	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap
Low	0.92	1.02	1.27	1.29	1.45	1.36	1.79	1.79	1.54
Medium	0.47	0.69	1.12	1.11	1.32	1.18	1.84	1.80	1.58
High	0.09	0.63	0.93	0.82	1.22	1.06	1.56	1.81	1.73
Low-high t -statistic	0.82* (3.65)	0.40 (1.56)	0.35 (1.46)	0.48* (2.76)	0.23 (1.12)	0.30 (1.61)	0.23 (1.17)	-0.02 (-0.10)	-0.19 (-0.83)

Mean Dispersion									
Dispersion	Losers						Winners		
	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap	Small Cap	Mid Cap	Large Cap
Low	0.10	0.05	0.03	0.04	0.03	0.02	0.03	0.02	0.02
Medium	0.21	0.10	0.06	0.09	0.05	0.04	0.06	0.15	0.04
High	1.18	0.85	0.55	0.61	0.33	0.22	0.39	0.28	0.25

^{a,b} Statistically significant at the one and five percent levels, respectively.

(a) Table V: size, momentum, and dispersion

Table VI
Time-Series Tests of Three- and Four-Factor Models for Dispersion Equal-Weighted Quintiles

This table reports estimates of the Fama-French three-factor model, as well as a four-factor model, $E(R_{it} - R_{ft}) = b_{1M}(R_{Mt} - R_{ft}) + s_i SML_t + h_i HML_t + m_i UMD_t$, for monthly excess returns on the equal-weighted dispersion quintiles. The market premium ($R_M - R_F$) uses the CRSP NYSE/AMEX/Nasdaq value-weighted index. The variables HML and SMB are created using the same methodology as Fama and French (1996). The momentum premium (UMD) is the difference between the return on a portfolio comprised of stocks with high returns from $t - 12$ to $t - 2$ and the return on a portfolio comprised of stocks with low returns from $t - 12$ to $t - 2$. The dispersion equal-weighted quintiles are formed as in Table II. The sample period is February 1983 to December 2000. Stocks with a price less than five dollars are excluded from the sample, and t -statistics (Newey-adjusted) are in parentheses.

Portfolio	Alpha (%)	Factor Sensitivities				Adj. R^2 (%)
		$R_M - R_F$	SMB	HML	UMD	
D1 (low-dispersion)	0.13 (0.88)	1.06 (39.78)	0.32 (3.17)	0.26 (2.83)		88.42
	0.27 (2.14)	1.06 (43.33)	0.33 (4.17)	0.21 (2.50)	-0.13 (-2.01)	89.18
	0.03 (0.30)	1.06 (44.13)	0.40 (5.76)	0.22 (2.95)		92.64
D2	0.17 (1.57)	1.06 (56.96)	0.41 (7.98)	0.17 (2.44)	-0.12 (-2.81)	93.35
	-0.07 (-0.82)	1.08 (42.71)	0.49 (6.59)	0.15 (2.91)		95.08
	0.09 (1.26)	1.07 (52.92)	0.51 (9.46)	0.09 (1.98)	-0.14 (-4.12)	95.94
D3	-0.18 (-2.56)	1.11 (52.79)	0.67 (12.74)	0.14 (3.31)		96.47
	-0.03 (-0.40)	1.10 (63.19)	0.68 (19.12)	0.08 (2.00)	-0.14 (-5.21)	97.17
	-0.58 (-4.78)	1.15 (38.53)	0.88 (16.16)	0.13 (3.19)		93.74
D4 (high-dispersion)	-0.35 (-2.62)	1.15 (45.66)	0.91 (26.12)	0.05 (1.11)	-0.20 (-5.18)	94.92
					GRS(3F) =	5.51
					GRS(4F) =	2.62

(b) Table VI: factor-model tests

6.7 Table VII: What predicts analyst forecast dispersion?

Read:

- Dispersion is positively related to beta, return volatility, earnings variability, turnover, leverage, and residual analyst coverage.
- Dispersion is negatively related to size and past returns.
- Since dispersion is positively related to many risk proxies, a risk explanation would naturally predict higher returns for high-dispersion stocks.
- The return evidence goes the other way: high-dispersion stocks earn lower returns.

Economic message: Dispersion captures uncertainty and disagreement, but not in a way that

looks like compensated risk. The market appears to overprice, rather than compensate, disagreement.

6.8 Table VIII: Subperiod analysis

Read:

- From 1983 to 1991, the all-stock low-minus-high spread is 1.16 percent per month and statistically significant.
- From 1992 to 2000, the all-stock spread falls to 0.41 percent and is not statistically significant.
- The small-stock spread remains positive and significant in the later period.
- Mean dispersion is lower in the later subperiod.

Economic message: The disagreement effect is stronger earlier in the sample, when trading costs and information frictions were plausibly higher. The persistence among small stocks fits the limits-to-arbitrage interpretation.

Table VII
FM Regressions of Dispersion in Analysts' Forecasts
on Lagged Firm Characteristics

Fama and Macbeth (1973) cross-sectional regressions are run every fiscal year from 1983 to 2000. Dispersion in analysts' forecasts (measured seven months before the end of the fiscal year) is regressed on $\hat{\beta}$ (estimated using the past 36 to 60 months of data), M (the lagged fiscal year-end market capitalization), B/M (the lagged fiscal year-end book-to-market ratio), R_{-12-2} (the return from $t-12$ months to $t-2$ months), $RCov$ (the residual from yearly regressions of $\ln(1 + \text{analyst coverage})$ on $\ln(M)$ and $\ln(B/M)$), $\sigma(R)$ (the standard deviation of daily returns over the past 250 days, lagged one month), Vol (the average adjusted daily volume from the last 250 days, lagged one month), $Turn$ (the average adjusted turnover from the last 250 days, lagged one month), D/B (the lagged fiscal year-end debt-to-book ratio), S/A (the lagged fiscal year-end sales-to-assets ratio), $\sigma(eps)$ (the one-year lagged standard deviation of earnings per share divided by the absolute value of mean earnings per share over the past five years), and Age (the mean age of the forecasts). A firm's adjusted volume is its volume divided by the mean volume of the firm's exchange, while a firm's adjusted turnover is its turnover divided by the mean turnover of the firm's exchange. Newey–West t -statistics are in parentheses. Stocks with a price below five dollars are excluded.

$\hat{\beta}$	$\ln M$	$\ln B/M$	R_{-12-2}	$RCov$	$\sigma(R)$	Vol	$Turn$	$\ln D/B$	$\ln S/A$	$\sigma(eps)$	Age
0.131 (4.49)											
	-0.046 (-6.59)										
		0.075 (3.26)									
			-0.309 (-5.22)								
				0.135 (4.86)							
					0.141 (4.95)						
						0.005 (2.99)					
							0.044 (8.20)				
								0.039 (3.00)			
									-0.034 (-4.25)		
										0.005 (2.71)	
											-0.023 (-1.37)
0.112 (3.81)	-0.025 (-7.19)	0.058 (2.66)	-0.282 (-5.40)	0.114 (5.67)							
-0.030 (-1.41)	0.005 (0.81)	0.089 (3.29)	-0.221 (-6.04)	0.087 (5.07)	0.185 (3.66)						
0.080 (2.51)	-0.028 (-6.98)	0.059 (2.53)	-0.295 (-5.78)	0.089 (4.33)		0.002 (1.79)	0.039 (3.42)				
0.067 (1.90)	-0.030 (-7.85)	0.032 (1.55)	-0.288 (-6.23)	0.088 (3.86)		0.002 (1.24)	0.046 (3.23)	0.039 (2.23)	-0.031 (-3.30)	0.004 (2.89)	-0.020 (-0.80)

Table VIII
Subperiod Analysis

Each month stocks are sorted in five groups based on the level of market capitalization as of the third Thursday of the previous month. Each size group is then sorted into five subgroups based on dispersion in analyst forecasts. Dispersion is defined as the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast, as reported in the I/B/E/S Summary History file. Stocks with a mean forecast of zero are assigned to the highest dispersion groups, and stocks with a price less than five dollars are excluded from the sample. Stocks are held for one month, and portfolio returns are equal-weighted. The table reports the average monthly return differential between stocks in the lowest- and highest-dispersion quintile in each size group over the indicated time periods; t -statistics in parentheses are adjusted for autocorrelation.

Mean Returns						
Time Period	Small				Large	All Stocks
	S1	S2	S3	S4	S5	
1983–2000	1.37 ^a (5.98)	0.89 ^a (3.12)	0.67 ^b (2.41)	0.48 (1.55)	0.29 (0.93)	0.79 ^a (2.88)
1983–1991	1.54 ^a (6.11)	1.41 ^a (5.28)	0.94 ^a (3.56)	0.86 ^b (2.62)	0.66 (1.95)	1.16 ^a (4.63)
1992–2000	1.21 ^a (3.14)	0.37 (0.77)	0.40 (0.83)	0.11 (0.21)	−0.08 (−0.16)	0.41 (0.86)
Mean Dispersion						
Dispersion Quintiles	Size Quintiles					All Stocks
	S1	S2	S3	S4	S5	
Time period: 1983–1991						
D1 (low)	0.013	0.014	0.016	0.018	0.017	0.015
D5 (high)	1.434	1.148	0.986	0.938	0.487	1.008
Time period: 1992–2000						
D1 (low)	0.008	0.008	0.009	0.010	0.010	0.009
D5 (high)	1.081	0.780	0.642	0.509	0.437	0.698

^{a,b} Statistically significant at the one and five percent levels, respectively.

(b) Table VIII: subperiod analysis

(a) Table VII: determinants of forecast dispersion

6.9 Figures 1 and 2: Timing robustness

Read:

- Figure 1 shows that the low-minus-high dispersion spread declines as the lag between the forecast signal and portfolio formation increases.
- With lags longer than roughly five months, the spread is no longer statistically significant.
- Figure 2 shows that the return spread remains positive for different holding periods, but declines as the holding period becomes longer.
- The patterns are consistent with a return signal that decays as information becomes stale and prices adjust.

Economic message: Disagreement predicts returns over relatively short horizons. The market correction is not instantaneous, but the information in forecast dispersion decays over time.

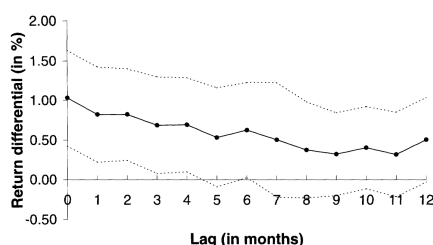


Figure 1. Robustness checks: Lags in portfolio formation. At the end of each month, all stocks in the I/B/E/S universe are ranked into deciles based on dispersion in analysts' earnings per share forecasts (defined as the ratio of the standard deviation to the absolute value of the mean forecast). Stocks in each dispersion decile are assigned into equal-weighted portfolios with a certain lag and held in the portfolio for one month. The difference in returns between the lowest- and highest-decile portfolios is calculated. The chart plots the mean monthly return differential, with the broken lines indicating the 95 percent confidence interval (adjusted for autocorrelation).

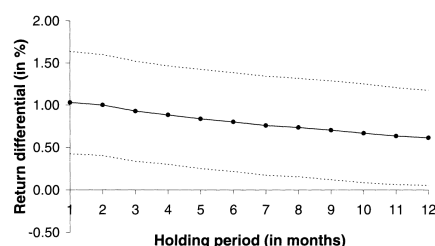


Figure 2. Robustness checks: Holding periods. At the end of each month, stocks are ranked into deciles based on dispersion in analysts' earnings per share forecasts and assigned into portfolios without a lag. The stocks are then held in the portfolio for T months, with $1/T$ th of each portfolio reinvested monthly. Portfolio returns are equal-weighted. The difference in returns between the lowest- and highest-decile portfolios is calculated. The chart plots the mean monthly return differential, with the broken lines indicating the 95 percent confidence interval (adjusted for autocorrelation).

(a) Figure 1: portfolio formation lags

(b) Figure 2: holding-period robustness

6.10 Table IX: Fama-MacBeth regressions explaining individual stock returns

Read:

- Table IX estimates monthly cross-sectional return regressions with standard controls.
- The coefficient on dispersion is negative in all specifications.
- The result survives controls for size, book-to-market, momentum, short-run reversal, long-run reversal, residual coverage, and turnover.
- Turnover does not subsume the forecast-dispersion effect.
- The standard deviation of forecasts is negative and significant, while the mean forecast is not the driver.

Economic message: The negative return effect of analyst disagreement is robust at the individual-stock level. It is not simply a portfolio-sorting artifact or a proxy for standard return predictors.

Table IX
Fama-MacBeth Regressions: Explaining the Cross Section of Individual Stock Returns
 Fama and MacBeth (1973) cross-sectional regressions are run every month from February 1980 to December 2000. The cross section of expected stock returns at time t is regressed on a constant (not reported), market β (estimated as in Fama and French (1992)), size (log of market capitalization at $t - 1$), book-to-market, residual coverage, turnover, the one-month lag of dispersion in analysts' forecasts, the one-month lag of standard deviation in analysts' forecasts, and the one-month lag of the mean analyst forecast. The (log) book-to-market ratio ($\ln(BE/ME)$) is computed by matching the yearly *BE* figure for all fiscal years ending in calendar year $t - 1$ to returns starting in July of year t ; this figure is then divided by market capitalization (*ME*) at month $t - 1$ to form the book-to-market ratio, so that the book-to-market ratio is updated each month. Residual coverage (*Resid. Cov.*) is the residual from monthly regressions of $\ln(1 + \text{analyst coverage})$ on $\ln(ME)$ and $\ln(BE/ME)$. Turnover (*Turn*) is the adjusted average turnover from the last 250 days (lagged one month); a firm's adjusted turnover is its turnover divided by the mean turnover of the firm's exchange. Dispersion (*Disp*) is the ratio of the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts to the absolute value of the mean forecast (observations with a mean forecast of zero are deleted from the sample). Standard deviation (*Std. Dev.*) is the standard deviation of analysts' current-fiscal-year annual earnings per share forecasts, and *Mean* is the mean analyst earnings forecast. The return on each stock from $t - 12$ to $t - 2$ is designated $ret_{-12:-2}$. In addition, $ret_{-1:-1}$ and $ret_{-36:-13}$ are included to capture the short-term and long-run reversal effects in individual stock returns. Stocks with a price less than five dollars are excluded from the sample. Statistics in parentheses are adjusted for autocorrelation.

β	$\ln(ME)$	$\ln(BE/ME)$	$ret_{-12:-2}$	$ret_{-1:-1}$	$ret_{-36:-13}$	Resid. Cov.	Turn	Disp	Std. Dev.	Mean
0.156	-0.039	0.065	0.013	-0.037	-0.002					
0.40	(-0.81)	(0.80)	(5.62)	(-6.25)	(-1.96)					
0.164	-0.040	0.069	0.013	-0.037	-0.002	0.008				
0.43	(-0.83)	(0.85)	(5.59)	(-6.27)	(-1.93)	(1.17)				
0.061	-0.033	0.094	0.013	-0.037	-0.002	0.012	-0.096			
0.18	(-0.69)	(1.10)	(5.84)	(-6.36)	(-1.51)	(1.70)	(-1.41)			
0.049	-0.036	0.094	0.013	-0.037	-0.002	0.012	-0.096	-0.094		
0.15	(-0.74)	(1.09)	(5.67)	(-6.36)	(-1.71)	(1.71)	(-1.43)	(-2.31)		
0.201	-0.031	0.108				0.014		-0.121		
0.46	(-0.63)	(1.22)				(1.74)		(-2.70)		
0.121	-0.025	0.116					-0.069	-0.118		
0.31	(-0.50)	(1.30)					(-0.96)	(-2.71)		
0.192	-0.029	0.107						-0.119		
0.43	(-0.61)	(1.20)						(-2.62)		
0.001	-0.045	0.109	0.012	-0.038	-0.002	0.011	-0.087		-0.447	0.042
0.00	(-0.95)	(1.29)	(5.39)	(-6.66)	(-2.04)	(1.53)	(-1.33)		(-2.07)	(1.19)
0.013	-0.039	0.153				0.016	-0.075		-0.649	0.060
0.04	(-0.82)	(1.74)				(2.11)	(-1.07)		(-2.93)	(1.49)
0.081	-0.041	0.147							-0.662	0.056
0.20	(-0.86)	(1.67)							(-2.76)	(1.31)

Table IX: Fama-MacBeth return regressions

7 Short summary

- The paper studies whether analyst forecast dispersion predicts future stock returns.
- Dispersion is defined as the cross-analyst standard deviation of EPS forecasts divided by the absolute value of the mean forecast.
- High-dispersion stocks earn lower future returns.
- The low-minus-high dispersion spread is large: about 0.79 percent per month in the baseline portfolio test.
- The effect is strongest among small stocks and past losers.
- Standard risk adjustments do not explain the result.
- Dispersion correlates positively with risk proxies, so the negative return relation is hard to square with a risk story.
- Robustness tests show that the effect is not driven by size, value, momentum, analyst coverage, turnover, or forecast scaling.
- The authors interpret dispersion as disagreement. When pessimistic opinions are constrained from affecting prices, high-disagreement stocks become overvalued.
- Analyst incentives may also censor pessimistic views, creating optimistic consensus forecasts when disagreement is high.

8 Conclusion

8.1 Core conclusion

Stocks with more disagreement among analysts earn lower future returns. The main empirical fact is not that disagreement raises expected returns as compensation for risk, but that disagreement

predicts underperformance.

8.2 Economic intuition

When investors disagree, the market price may be set by relatively optimistic investors. Pessimists may not short because shorting is costly, risky, or institutionally constrained. Similarly, pessimistic analysts may avoid issuing very negative forecasts. As a result, negative information is muted and prices can be biased upward.

8.3 Incremental contribution revisited

The paper turns a classic theoretical idea - optimistic pricing under heterogeneous beliefs and short-sale constraints - into a large-sample empirical asset-pricing test. It also shows that forecast dispersion is not just a risk proxy, because its return relation is negative and survives standard risk adjustment.

8.4 Implications for investors

High analyst disagreement is a warning sign. A high-dispersion stock may look attractive to optimists, but the average future return evidence suggests that disagreement often reflects overpricing rather than an opportunity for higher compensation.

8.5 Implications for asset pricing

The paper supports models in which heterogeneous beliefs and limits to arbitrage matter for prices. Standard representative-agent or fully arbitrated models have difficulty explaining why disagreement predicts low returns.

8.6 Limitations

The paper does not directly observe every investor's beliefs or every short-sale constraint. Analyst dispersion is a proxy for disagreement, not disagreement itself. The sample is also limited to stocks covered by at least two analysts, so the results may not describe uncovered microcaps or severely distressed firms outside I/B/E/S.

8.7 Takeaway

The central takeaway is that disagreement is not innocuous. In the cross-section of stocks, high dispersion in analysts' earnings forecasts identifies stocks whose prices appear too optimistic and whose subsequent returns are low.